

PART I - FOUNDATION PLC PROJECTS

Project 3 - Automatic Water Tank Level Control System

Version 1 - Foundation PLC Project

PLC Platform: Allen-Bradley MicroLogix 1100

Programming Software: RSLogix Micro Starter Lite 8.30

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1. Project Overview

The Automatic Water Tank Level Control System was developed to introduce level-based process control using latch/unlatch logic and timer-based runtime monitoring. The objective was to automatically start a pump when the tank reached a low-level condition and stop the pump when the high-level condition was reached.

The system uses Low Level Sensor I:1/0 to latch Pump output O:2/0 ON and High Level Sensor I:1/1 to unlatch the same output OFF. This allows the pump to continue running after the low-level condition clears and remain active until the high-level condition is detected.

A separate TON timer T4:0 monitors pump runtime. If the pump remains ON continuously until the timer reaches its 30-second preset, Alarm Lamp O:2/1 is energized.

This project introduced several important PLC concepts at the same time: latched output control, two-level process operation, timer supervision, and alarm indication.

2. Engineering Objective

The objective of this project was to develop and verify a PLC program that automatically controlled a water pump between low-level and high-level conditions while also monitoring excessive pump runtime.

- Start the pump when the low-level condition was detected.
- Keep the pump running after the low-level signal cleared.
- Stop the pump when the high-level condition was detected.
- Monitor the continuous pump runtime using a TON instruction.
- Turn on an alarm indication when the runtime timer reached 30 seconds.

This project also gave me practical experience using OTL, OTU, TON, timer status bits, and separate output logic.

3. Functional Requirements

- The PLC shall monitor Low Level Sensor I:1/0.
- When I:1/0 becomes true, Pump output O:2/0 shall be latched ON.
- The pump shall remain ON after the low-level input returns false.
- The PLC shall monitor High Level Sensor I:1/1.
- When I:1/1 becomes true, Pump output O:2/0 shall be unlatched OFF.
- Timer T4:0 shall accumulate while Pump O:2/0 is ON.
- T4:0 shall use a 0.01-second time base and a preset of 3000.
- The resulting runtime threshold shall be 30 seconds.
- Alarm Lamp O:2/1 shall remain OFF while T4:0/DN is false.
- Alarm Lamp O:2/1 shall energize when T4:0/DN becomes true.
- When the pump turns OFF, the TON shall reset automatically.
- Resetting the timer shall clear the Done bit and de-energize the Alarm Lamp.

4. System Components

Component	Identification	Function
PLC Platform	Allen-Bradley MicroLogix 1100	Executes the level-control program
Input Module	1762-IQ16	Provides discrete level-sensor inputs
Output Module	1762-OB16	Provides pump and alarm outputs
Low Level Sensor	I:1/0	Requests pump operation when the tank is low
High Level Sensor	I:1/1	Stops pump operation when the tank reaches high level
Pump	O:2/0	Controls the simulated pump command
Runtime Timer	T4:0	Monitors continuous pump runtime
Alarm Lamp	O:2/1	Indicates that the 30-second runtime threshold has been reached
Programming Software	RSLogix Micro Starter Lite 8.30	Ladder logic development and monitoring

The Version 1 program models the control logic only. It does not include a real motor starter, overload device, safety circuit, analog level transmitter, or physical process instrumentation.

5. I/O Assignment

Physical I/O

Device	PLC Address	Type	Description
Low Level Sensor	I:1/0	Digital Input	Detects low tank level
High Level Sensor	I:1/1	Digital Input	Detects high tank level
Pump	O:2/0	Digital Output	Pump control command
Alarm Lamp	O:2/1	Digital Output	Excessive-runtime indication

Internal PLC Data

Function	Address	Type	Purpose
Runtime Timer	T4:0	TON	Monitors continuous pump runtime
Timer Done	T4:0/DN	Status Bit	Becomes true at the 30-second threshold
Time Base	-	0.01 second	Timer resolution
Preset	-	3000	Runtime threshold

The resulting runtime threshold is: $3000 \times 0.01 \text{ second} = 30.00 \text{ seconds}$.

6. Control Strategy

I used a four-function control strategy: pump start, pump stop, runtime monitoring, and alarm indication.

Low Level Sensor I:1/0 executes an OTL instruction on Pump O:2/0. High Level Sensor I:1/1 executes an OTU instruction on the same pump output. When Pump O:2/0 is ON, T4:0 begins timing. When T4:0/DN becomes true, Alarm Lamp O:2/1 energizes.

This approach allows the pump state to be maintained independently of the low-level sensor after startup. Once latched, the pump continues running until the high-level condition explicitly removes the latched state.

The timer does not control the pump directly. It supervises how long the pump has remained ON and provides an indication when the configured runtime threshold is reached.

7. Engineering Design Decisions

Design Decision	Reason
Used OTL O:2/0 for pump start	The pump needed to remain ON after the low-level signal cleared
Used OTU O:2/0 for pump stop	The high-level condition needed to explicitly remove the latched pump state
Used separate low and high level sensors	This created a defined operating band between pump start and stop conditions
Used TON T4:0 for runtime monitoring	The application required continuous pump-runtime supervision
Set timer to 3000 at a 0.01-second time base	This created the required 30-second threshold
Used T4:0/DN to control the alarm	The Done bit directly represents that the runtime threshold has been reached
Kept the alarm as an OTE output	Version 1 only required an indication, not a latched fault or shutdown

One important limitation is that the pump is latched and unlatched directly through the physical output address O:2/0. I retained this exactly as implemented in Version 1. In the later industrial version, I would prefer to latch an internal fill request or run request and control the physical pump output separately through permissives and interlocks.

8. Ladder Logic Description

The Version 1 program contains four functional rungs. Each rung is shown below using the approved single-line ladder format and then explained based on the actual RSLogix implementation.

8.1 Rung 0000 - Start / Latch Pump

```
|----[ XIC I:1/0 ]----- ( OTL O:2/0 )----|
      Low_Level_Sensor           Pump
```

When Low Level Sensor I:1/0 becomes true, the OTL instruction latches Pump output O:2/0 ON. After the pump has been latched, the low-level sensor does not need to remain active for the pump to continue running.

8.2 Rung 0001 - Stop / Unlatch Pump

```
|----[ XIC I:1/1 ]----- ( OTU O:2/0 )----|
      High_Level_Sensor           Pump
```

When High Level Sensor I:1/1 becomes true, the OTU instruction unlatches Pump O:2/0. This stops the filling operation when the high-level condition is reached.

8.3 Rung 0002 - Pump Runtime Monitoring

```
|----[ XIC O:2/0 ]-----[ TON T4:0 ]----|
      Pump                Run_time_Timer  PRE: 3000  Time Base: 0.01
```

When Pump O:2/0 is ON, Timer T4:0 begins accumulating. If the pump turns OFF before 30 seconds, the non-retentive timer resets.

8.4 Rung 0003 - Runtime Alarm Indication

```
|----[ XIC T4:0/DN ]----- ( OTE O:2/1 )----|
      Run_time_Timer_DN           Alarm_Lamp
```

When T4:0 reaches its preset, T4:0/DN becomes true and energizes Alarm Lamp O:2/1. The alarm is not latched. If the pump turns OFF, T4:0 resets, the Done bit clears, and the Alarm Lamp turns OFF automatically.

8.5 Actual RSLogix Implementation

The clean ladder representations above are included for readability. The figure below shows the corresponding ladder logic as implemented in RSLogix Micro Starter Lite.

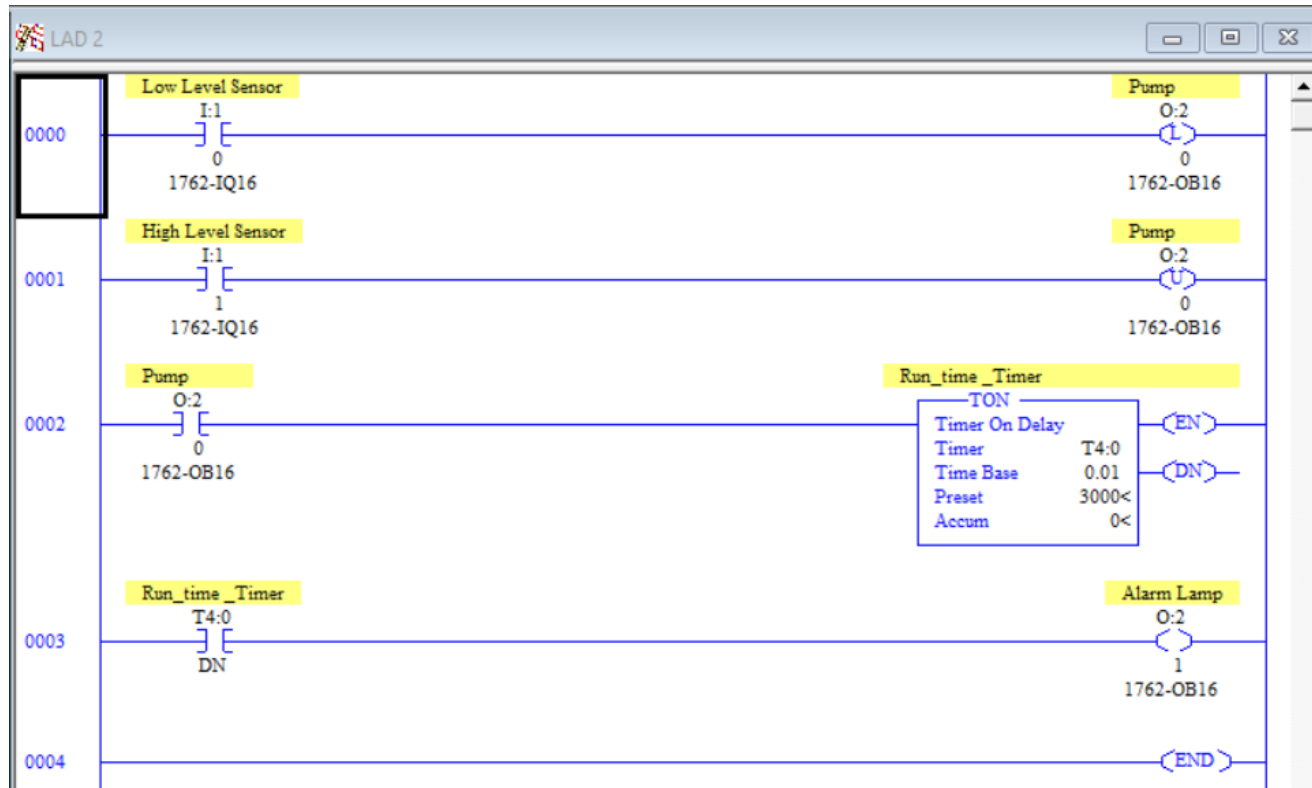


Figure P3.1 - RSLogix Implementation of the Version 1 Automatic Water Tank Level Control System

9. Sequence of Operation

Step 1 - Initial Condition: The pump is OFF, runtime timer T4:0 is reset, and the Alarm Lamp is OFF.

Step 2 - Low Level Detected: Low Level Sensor I:1/0 becomes true.

Step 3 - Pump Starts: OTL O:2/0 latches the pump ON.

Step 4 - Tank Fills: The pump remains ON while the tank level rises.

Step 5 - Runtime Monitoring Begins: Because Pump O:2/0 is ON, T4:0 begins accumulating.

Step 6A - High Level Reached Before 30 Seconds: High Level Sensor I:1/1 becomes true and unlatches the pump. The timer resets and the Alarm Lamp remains OFF.

Step 6B - Pump Continues for 30 Seconds: If the pump remains ON until T4:0 reaches its preset, T4:0/DN becomes true and Alarm Lamp O:2/1 turns ON.

Step 7 - High Level Reached: The high-level input unlatches the pump.

Step 8 - Timer and Alarm Reset: When the pump turns OFF, the TON resets. The Done bit clears and the Alarm Lamp turns OFF.

10. Testing & Validation

I tested the completed ladder logic in RSLogix Micro Starter Lite by monitoring the level inputs, pump output, timer status, and alarm output under the defined operating conditions.

Test ID	Test Action	Expected Result	Observed Result	Status
T-01	Activate Low Level Sensor I:1/0	Pump O:2/0 latches ON	Pump turned ON and remained latched	PASS
T-02	Activate High Level Sensor I:1/1	Pump O:2/0 unlatches OFF	Pump turned OFF	PASS
T-03	Run pump for less than 30 seconds	T4:0/DN remains false and Alarm Lamp stays OFF	Alarm remained OFF	PASS
T-04	Run pump continuously until timer reaches 30 seconds	T4:0/DN becomes true and Alarm Lamp turns ON	Alarm energized at the runtime threshold	PASS

Validation Summary

Testing confirmed that the pump started correctly from the low-level condition and remained latched after startup.

The high-level condition successfully removed the latched pump state and stopped the pump.

While the pump runtime remained below 30 seconds, the Alarm Lamp stayed OFF. When the pump remained ON until the timer reached its preset, T4:0/DN became true and the Alarm Lamp energized.

All four defined Version 1 functional tests passed.

The validation was performed within the project development environment and represents functional PLC program validation, not industrial commissioning or safety validation.

11. Results

The completed Automatic Water Tank Level Control System successfully performed the required pump start, pump stop, runtime monitoring, and alarm indication functions.

The low-level condition latched the pump ON, while the high-level condition unlatched it OFF. This provided a simple two-level control sequence in which the pump remained active between the low and high level points.

Timer T4:0 successfully monitored continuous pump runtime. The alarm remained OFF below the 30-second threshold and energized when the timer reached its preset.

All four defined functional tests passed.

12. Challenges & Troubleshooting

The most useful troubleshooting path for this project was: Low Level / High Level Inputs -> Pump Latched State -> Runtime Timer -> Timer Done Bit -> Alarm Lamp.

Symptom	Possible Cause	Diagnostic Check	Corrective Action
Pump does not start	Low Level Sensor not active or latch rung not true	Monitor I:1/0 and Rung 0000	Verify input condition and OTL O:2/0
Pump does not stop	High Level Sensor not active or unlatch rung not true	Monitor I:1/1 and Rung 0001	Verify high-level condition and OTU O:2/0
Pump state behaves unexpectedly	Latched output state not understood or conflicting conditions occur	Monitor O:2/0, Rung 0000, and Rung 0001	Verify latch/unlatch sequence and input states
Timer does not run	Pump output is not ON	Monitor O:2/0 and T4:0	Verify pump command before

			reviewing timer logic
Alarm never turns ON	Timer preset/configuration or Done bit is incorrect	Monitor T4:0.ACC and T4:0/DN	Verify timer configuration and Rung 0003
Alarm remains ON unexpectedly	Pump remains latched ON or timer has not reset	Monitor O:2/0, T4:0/DN, and high-level input	Verify pump unlatch condition and timer reset

One of the key lessons from this project was that latch/unlatch logic requires careful attention to the current stored state of the output. Unlike an OTE, the output does not automatically follow only the current rung condition.

13. Lessons Learned

This project helped me understand how latch and unlatch instructions can be used to maintain a control state between two separate process conditions.

I learned that the low-level sensor could initiate pump operation without needing to remain continuously true, while the high-level sensor could explicitly remove the latched state when the required tank level was reached.

The project also reinforced the use of timers for supervision rather than direct control. T4:0 did not start or stop the pump; instead, it monitored the duration of pump operation and provided an alarm condition when the configured runtime threshold was reached.

I also learned that an alarm indication and a fault shutdown are not the same thing. In this Version 1 project, the alarm only provides an indication and does not stop the pump automatically.

14. Industrial Relevance

Automatic tank-level control is widely used in municipal water systems, water treatment plants, chemical storage tanks, boiler feed-water systems, cooling towers, and fire-water storage tanks.

The Version 1 logic represents the basic control concept of starting a pump at a low level and stopping it at a high level.

In a real industrial system, additional considerations would normally include pump overload protection, dry-run protection, sensor plausibility checking, pump permissives, Manual/Auto operation, alarm acknowledgement, fault latching, high-high or low-low level protection, redundant instrumentation where required, process interlocks, and maintenance/restart behavior.

The Version 1 project therefore serves as a foundation for a more complete industrial tank-control system.

15. Transition to Industrial Version (Version 2)

The Version 2 design can retain the same basic low-level/start and high-level/stop requirement while improving the structure and diagnostic capability of the system.

Version 2 Improvement	Engineering Benefit
Internal Fill Request / Pump Run Request	Separates latched process state from the physical pump output
Auto/Manual operating modes	Supports normal operation and maintenance control
Pump permissives	Prevents pump operation unless required conditions are satisfied
Motor overload feedback	Detects and responds to motor protection trips
Dry-run protection	Prevents pump operation when suction/process conditions are unsafe
Sensor conflict detection	Identifies impossible or inconsistent low/high sensor states
High-high level alarm	Provides independent overflow warning

Low-low level alarm	Provides additional low-level diagnostic or protection
Latched runtime fault	Preserves abnormal runtime condition until reset
Fault reset logic	Provides controlled recovery after abnormal conditions
HMI tank/pump status	Gives operators clear level, pump, timer, alarm, and fault information
Runtime diagnostics	Helps identify slow filling, blocked flow, failed pump, or process problems
Pump command separation	Allows the physical output to be controlled through permissives, interlocks, and faults

One of the biggest improvements will be replacing the direct OTL/OTU control of O:2/0 with an internal operating request.

Low Level Detected -> Fill Request Latched -> Sensor / Permissive / Fault Checks -> Pump Command -> Runtime Supervision -> High Level Reached -> Fill Request Cleared -> Pump Stops

If an abnormal runtime condition occurs, Version 2 can decide whether to latch a fault, stop the pump, require operator reset, or provide additional diagnostics according to the final control philosophy. This creates a clear engineering progression from simple level control to a more structured industrial pump-control architecture.

16. Project Summary

The Automatic Water Tank Level Control System introduced level-based process control using latch/unlatch logic and runtime supervision.

Low Level Sensor I:1/0 latched Pump O:2/0 ON, while High Level Sensor I:1/1 unlatched the pump OFF. This allowed the pump to remain active between the low and high level conditions without requiring the low-level input to stay continuously active.

Timer T4:0 monitored pump runtime and energized Alarm Lamp O:2/1 when the 30-second runtime threshold was reached. All four defined functional tests passed.

The project established my understanding of maintained PLC states, two-level process control, timer-based supervision, and alarm indication. It also provides a clear foundation for Version 2, where the system can be expanded with internal run requests, sensor validation, permissives, fault handling, operator modes, and improved diagnostics.